

BUS63-4

Strategic Allocation

Committing scarce resources when prediction is cheap and judgment is the binding constraint

The Allocate stage of the Strategy arc: deploying capital, attention, and judgment under irreversibility.

Albert School — 22 hours

Preface

This is the final movement of the Strategy arc — Build → Compete → Evolve → **Allocate** — and you arrive carrying the whole arc on your back. From **BUS23-2** you can decode a business as a designed object: its unit economics, its defensibility, the gap between the value it *creates* and the value it *captures*. From **BUS33-2** you can reason strategy against strategy — Five Forces, VRIO, generic strategies, activity systems, and the failure modes of imitation, drift, and unjustified diversification. From **BUS23-3** you can read a firm’s financial vital signs: the DuPont decomposition, ROIC, and the free cash flow that decisions actually turn on. From **BUS33-3** you can price time and risk — NPV, WACC, the value-creation test ROIC versus WACC. And from **BUS43-3** you can value a firm under hypothesis sensitivity and reason about irreversible decisions with the intuition of real options.

What changes here is the **altitude** of the question. Until now you have analysed one business, one rivalry, one valuation. Now you sit where the capital, the attention, and the talent of an entire organisation are committed — and you ask the leader’s question: *given that these resources are scarce and the commitments are largely irreversible, where do they go?* BUS33-2 forced strategy against strategy under constraint; this course poses what comes after the strategy is chosen — how to back it with resources, coherently, when you cannot fund everything and cannot easily undo what you fund.

This course shares its central question — capital allocation — with **BUS63-1 Corporate Finance Leadership**, but approaches it from a different chair. CFO Leadership runs allocation *from inside the finance function*: profitability indexes, covenant headroom, the mechanics of payout. Strategic Allocation takes the **general-management posture**: the same euro, judged not only on its return but on its coherence with the firm’s declared strategy, its effect on future flexibility, and its exposure to rivals. The two courses meet at ROIC and diverge in everything around it.

The conceptual anchor of the course is the framework of Agrawal, Gans, and Goldfarb: **artificial intelligence makes prediction cheap, and so human judgment — the complement to prediction — becomes the binding scarce asset**. That single shift reorganises the whole problem of allocation. When prediction was expensive, much of management was the gathering and processing of information. When prediction is cheap, the scarce inputs are capital (where to commit it), attention (which decisions deserve it), and judgment (which decisions a human must own). This book is the discipline of committing those three scarce assets coherently.

One firm runs through the whole book to keep the ideas concrete. **Helios Group** is a fictional mid-cap conglomerate with three businesses: a mature, cash-generative *Industrial* arm; a faster-growing *Logistics* platform; and a small *Data Services* unit experimenting with AI at scale. Helios earns about €4{,}000m of revenue, holds €5{,}000m of invested capital, and faces a 9% cost of capital. Watch how one set of facts forces connected choices: where Helios deploys capital constrains what it can build versus buy, which shapes how it must organise decisions, which determines where human judgment must be preserved. That web of linked commitments *is* strategic allocation, and the course terminates in the deliverable that tests it whole: a strategic allocation audit of a real organisation, defended before a jury.

Contents

Preface	2
1 ROIC as a strategic allocation compass	3
1.1 The compass, made concrete	3

1.2	Diagnosing value-destroying growth	4
1.3	The limits of the compass	5
2	Make, buy, or partner — and the shape of the firm	5
2.1	Three modes, and why cost is not the criterion	5
2.2	Corporate scope: which businesses, and the discipline of exit	6
3	M&A as a strategic instrument	7
3.1	When to acquire — and the three failure mechanisms	7
3.2	Diagnosing a real capital allocation decision	7
4	Executive attention as a scarce resource	8
4.1	Simon’s economics of attention	8
4.2	Centralisation, delegation, and overload	8
4.3	Analysing the decision architecture	9
5	AI as cheap prediction: the Agrawal–Gans–Goldfarb framework	9
5.1	Prediction and judgment	9
6	a prediction (what will happen?) combined with a judgment (what do we want, and	10
6.1	Classifying decisions	10
6.2	Defending the human-judgment boundary	11
7	Organisational redesign for the AI era	11
7.1	Reconfiguring decision structures	11
7.2	New sources of managerial leverage	12
8	Measurable risk versus deep uncertainty	12
8.1	The distinction, and the rule it dictates	12
8.2	Robust versus optimised strategies	13
9	Real-options reasoning	13
9.1	The value of flexibility	13
9.2	Commit versus preserve optionality	14
9.3	Irreversibility and explicit exit conditions	14
10	The strategic allocation audit	15
10.1	Coherence is the deliverable	15
10.2	What the deliverable must carry	15

1 ROIC as a strategic allocation compass

Helios’s Logistics arm grew revenue 30% over three years, and the management team celebrated. Yet over the same period the group’s share price drifted sideways. How can a business grow a third larger and create no value? The answer is the single most important diagnostic in this course, and it is not a growth number at all. It is the spread between what the firm earns on its capital and what that capital costs.

1.1 The compass, made concrete

You computed Return on Invested Capital in BUS23-3 and tested it against WACC in BUS33-3. Here ROIC stops being a diagnostic endpoint and becomes a *compass*: the instrument that tells a leader whether any deployment of capital — a new factory, an acquisition, a geographic entry — points toward value or away from it.

Definition 1 (ROIC and the value-creation test) Return on invested capital is the after-tax operating return the firm earns on the capital put to work in its business, independent of how that capital is financed:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}} = \frac{\text{EBIT} \times (1 - t)}{\text{Invested capital}}$$

Capital allocation creates value when, and only when, the firm earns more on its invested capital than that capital costs:

$\text{ROIC} > \text{WACC} \Rightarrow \text{value created}$, $\text{ROIC} = \text{WACC} \Rightarrow \text{value-neutral}$, $\text{ROIC} < \text{WACC} \Rightarrow \text{value destroyed}$.

The value created per year is the *spread* times the capital deployed: $(\text{ROIC} - \text{WACC}) \times \text{Invested capital}$.

Worked example — Helios Logistics — growth that destroyed value. Logistics deployed €1{,}000m of incremental capital to win its 30% revenue growth, but the new contracts earn a 6% ROIC against the group’s 9% WACC. The arithmetic is brutal: $(0.06 - 0.09) \times 1000 = -\text{€}30\text{m}$ of value destroyed *every year*, even as revenue, headcount, and warehouse count all climb. Growth was not the cure for Helios’s flat share price — growth was the *mechanism* of the damage. The same capital left in Industrial, which earns 14%, would have created $(0.14 - 0.09) \times 1000 = \text{€}50\text{m}$ a year.

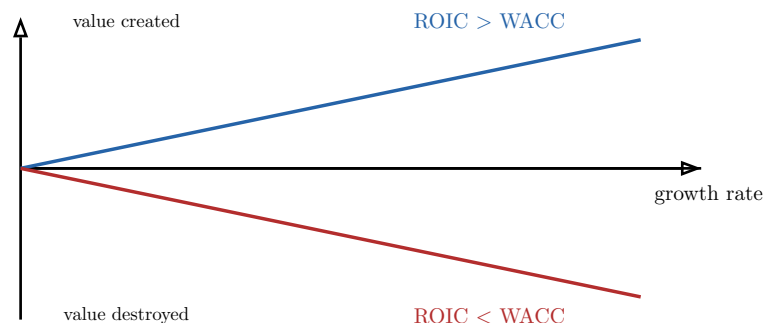


Figure 1: The allocation compass. When ROIC exceeds WACC, faster growth creates more value (blue); when ROIC is below WACC, faster growth destroys *more* value (red) — the firm scales a money-losing machine. Revenue growth alone tells you nothing; only the ROIC–WACC spread does.

1.2 Diagnosing value-destroying growth

The most dangerous failure mode is the one that looks like success. To diagnose whether reported growth destroys value:

1. Compute ROIC and WACC on the *same* invested-capital base.
2. Compare them. If $\text{ROIC} < \text{WACC}$, every incremental euro of growth consumes value.
3. Separate **accounting growth** (revenue, earnings, assets rising) from **returns above the cost of capital**. The two routinely move in opposite directions, and a leader who watches only the first is flying blind.

Common pitfall Revenue growth is value-neutral until you know the ROIC behind it. The remedy for value-destroying growth is almost never “grow less” in the abstract — it is “raise the ROIC of what you are growing, or stop pouring capital into it.” A leader who responds to a low-ROIC business by demanding *more* revenue growth is accelerating toward the cliff, not away from it.

1.3 The limits of the compass

A compass points; it does not measure distance. ROIC is a backward-looking accounting number, and three limits matter for allocation:

- **Definitional sensitivity.** The figure depends on how invested capital is defined and on the comparability adjustments of BUS23-3 — R&D capitalisation, lease treatment, stock-based compensation. Inconsistent definitions make cross-firm or cross-segment comparison meaningless.
- **Timing.** Capital deployed today may earn its return only in later periods, so a single-year ROIC understates the return on long-gestation investments. A young business with a low current ROIC is not automatically a value-destroyer — it may be early.
- **It judges the past, not the future.** ROIC tells you what the existing capital base earned. Allocation is about the *next* euro, which requires the forward-looking tools of the later chapters — real options especially.

Remark This is the same value-creation test BUS63-1 uses as the CFO's compass. The difference in posture is what you do with the reading: the CFO rations capital by profitability index; the general manager asks whether a sub-WACC business should exist in the portfolio at all — a question of corporate scope, which is where we turn next.

2 Make, buy, or partner — and the shape of the firm

ROIC tells you *whether* to commit capital to a capability. It does not tell you *how* to obtain it, or *which businesses* the firm should be in at all. Those are the two scope questions of this chapter: the mode decision (make/buy/partner) and the boundary decision (what is in the portfolio).

2.1 Three modes, and why cost is not the criterion

Worked example — Helios needs a cold-chain capability. Logistics wants to carry temperature-controlled freight. It can **make** — build refrigerated depots and hire the expertise (full control and IP, but two years and uncertain execution); **buy** — acquire a regional cold-chain operator (capability next quarter, but a steep control premium and integration risk); or **partner** — sub-contract to a specialist under a long-term contract (cheap and fast, but the capability is rented, and a rival can rent the same one tomorrow). The cheapest option, partnering, may be the strategically worst if cold-chain is meant to be a source of advantage — because it leaves Helios undifferentiated and exposed.

Definition 2 (Make, buy, or partner) To obtain a strategic capability or enter an activity, a firm can:

- **make** — build it internally (most control and commitment, slowest, highest capital);
- **buy** — acquire it (fastest to scale, full exposure, integration risk — treated as M&A in Chapter 3);
- **partner** — access it through an alliance, joint venture, or contract (least capital and control, preserves flexibility, creates dependence).

The decision is *not* reducible to which mode is cheapest. At corporate scale it is weighed on three criteria, and cost is at most a constraint:

- **strategic coherence** — does the activity fit the firm’s existing activity system and declared strategy (BUS33-2)? A capability that is core to the strategy argues for *make* a peripheral one argues for *partner*.
- **flexibility** — does the mode preserve or foreclose future options (Chapter 8)? Partnering keeps options open; making commits them.
- **competitive exposure** — does the mode create dependence on, or hand advantage to, a rival, or a partner who may later become one?

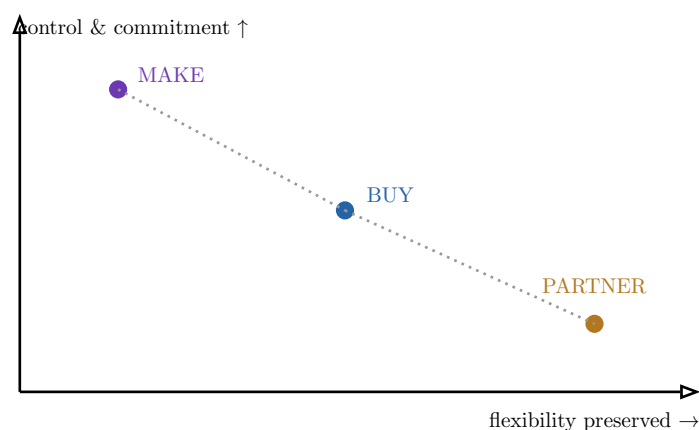


Figure 2: The mode trade-off. Making maximises control and commitment but consumes flexibility; partnering preserves flexibility but cedes control; buying sits between. The right mode depends on how core the capability is — not on which is cheapest this year.

2.2 Corporate scope: which businesses, and the discipline of exit

Scope is the set of businesses, geographies, and activities the firm chooses to be in. The scope levers in this course are **business mix** (which businesses), **geography** (which markets), **vertical integration** (which stages of the value chain to own versus buy — a make/buy decision along the chain), **portfolio management** (allocating capital and attention across the mix), and **divestiture discipline** (exiting what no longer earns its keep).

Definition 3 (Divestiture discipline) The practice of exiting a business when it ceases to create value (ROIC below WACC with no credible path back) or to cohere with the strategy — rather than retaining it by default. Its absence is one of the most common sources of value-destroying scope: businesses are added eagerly and shed reluctantly, so conglomerates accrete sub-WACC units that no one will champion killing.

Worked example — Helios's portfolio review. Helios’s portfolio review finds Data Services earning 4% on €300m of capital against a 9% WACC, with no line of sight to the spread closing. Two defensible moves exist: fix it (a credible plan to raise ROIC above 9%) or exit it (sell or wind down, redeploying the €300m to Industrial’s 14% projects). The *indefensible* move — and the default one — is to keep funding it because it is “strategic,” “nearly there,” or simply nobody’s job to close. Divestiture discipline is the willingness to make exit a live option, not a last resort.

Common pitfall Unjustified diversification (BUS33-2’s failure mode) reappears here as a scope error. Adding a business is justified only if the firm can earn above its cost of capital *in that business* and the business coheres with the rest. “It is a growth market” is not a

justification — a high-growth business in which the firm has no advantage will earn sub-WACC returns and destroy value faster the more it grows (Chapter 1). Scope must be earned on the compass, not on the size of the addressable market.

3 M&A as a strategic instrument

Acquisition is the “buy” mode, but it deserves its own chapter because it is the allocation decision that destroys the most value most reliably — and the one leaders are most tempted to over-use. The empirical record is sobering: a large share of acquisitions destroy value *for the acquirer*, even as they enrich the target’s shareholders.

3.1 When to acquire — and the three failure mechanisms

Definition 4 (The acquirer's value equation) An acquisition creates value for the acquirer only if the value of the realisable synergies exceeds the control premium paid over the target’s standalone value:

$$\text{value to acquirer} = \text{synergies realised} - \text{control premium paid.}$$

Because the premium is paid *up front and with certainty* while synergies are *future and uncertain*, the equation is biased against the acquirer before integration even begins.

The recurring mechanisms of deal value destruction are:

- **overpayment** — paying a control premium that exceeds the value of realisable synergies, often driven by competitive bidding or executive ego;
- **overestimated synergies** — synergies that prove smaller, later, or costlier to realise than the deal model assumed;
- **integration failure** — the combined organisation cannot capture the value the deal assumed, because cultures, systems, or customers do not merge as planned.

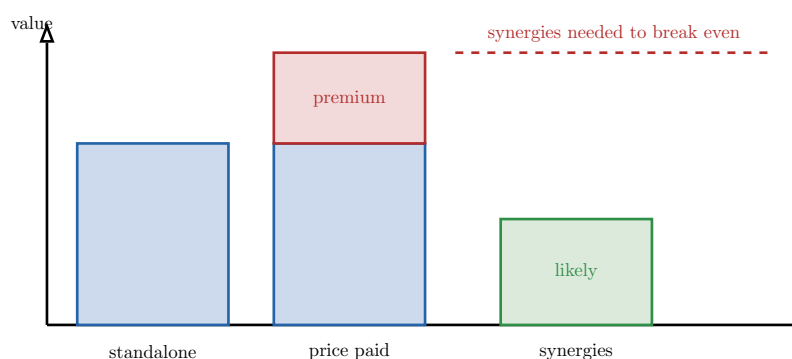


Figure 3: The acquirer’s gap. The premium (red) is paid with certainty; to break even, synergies must equal the premium (dashed line), yet realised synergies (green) typically fall short. Value transfers from acquirer to target shareholders — the empirical norm, not the exception.

3.2 Diagnosing a real capital allocation decision

Learning Outcome 3 asks you to judge a reported large-firm decision — an acquisition, divestiture, buyback, or organic investment — as value-creating or value-destroying. The method is the same regardless of the instrument:

1. **State the decision and the capital committed** — what was done, for how much.

2. **Name the causal mechanism** by which value is created or destroyed — e.g. a control premium exceeding synergies; a buyback executed *above* intrinsic value; organic investment earning below WACC; a divestiture sold *below* the value the buyer then extracted.
3. **State the counterfactual** — what the firm would have done with the same capital instead, and whether that alternative would have created more value. A judgment without a counterfactual is an opinion, not a diagnosis.

Worked example — Reading a buyback. Suppose Helios buys back €500m of its own shares. This creates value only if the shares were trading *below* intrinsic value — then the firm retires cheap equity and concentrates value in the remaining holders. If the shares were *above* intrinsic value, the buyback destroys value as surely as overpaying for an acquisition: the firm is “acquiring” its own overpriced stock. The counterfactual — funding Industrial’s 14% projects with that €500m — would have created $(0.14 - 0.09) \times 500 = \text{€}25\text{m}$ a year. The buyback is justified only if no such project existed and the shares were genuinely cheap.

Common pitfall Do not judge a deal by whether the share price rose afterward — markets are noisy and the move may reflect everything except the deal. Judge it by the mechanism and the counterfactual. Conversely, a deal that “felt strategic” and that management defended eloquently can still be a value destroyer if the premium exceeded the synergies. The instrument (acquire, build, partner) must be selected against coherence, flexibility, and exposure — never on the availability or apparent cheapness of a target on offer.

4 Executive attention as a scarce resource

Capital is the obvious scarce resource. The subtler one — and the one that silently caps the quality of every decision in a large firm — is the attention of the people who must decide. Herbert Simon saw this before the information age made it acute.

4.1 Simon’s economics of attention

Definition 5 (The economics of attention) “A wealth of information creates a poverty of attention.” When information is abundant, the binding constraint on decision-making is not data but the limited attention of those who must process it and decide. Executive attention is therefore a scarce input to be *allocated*, exactly like capital.

Worked example — Helios’s CEO and the 200 decisions. Helios’s CEO chairs an investment committee that approves every commitment above €5m. Last year that was 200 decisions. The first twenty got days of scrutiny; the last hundred got minutes each, rubber-stamped because the committee was saturated. The scarce resource was never the firm’s capital — Helios had cash — it was the committee’s *attention*. By routing trivial and major decisions through the same bottleneck, Helios spent its scarcest asset on the wrong things.

4.2 Centralisation, delegation, and overload

Because attention is scarce, decisions must be distributed across the hierarchy. This is the central trade-off:

- **Centralisation** concentrates decisions at the top — preserving coherence and firm-wide perspective, but consuming scarce executive attention and creating bottlenecks.

- **Delegation** pushes decisions down — conserving executive attention and exploiting local information, but risking loss of coherence across the firm.

Definition 6 (Hierarchy overload) Hierarchy overload occurs when more decisions are routed to a level of the hierarchy than its attention capacity can process with quality — degrading the quality of *every* decision at that level, not just the marginal one. Adding one more decision to a saturated executive does not get a worse decision; it makes all of them worse.

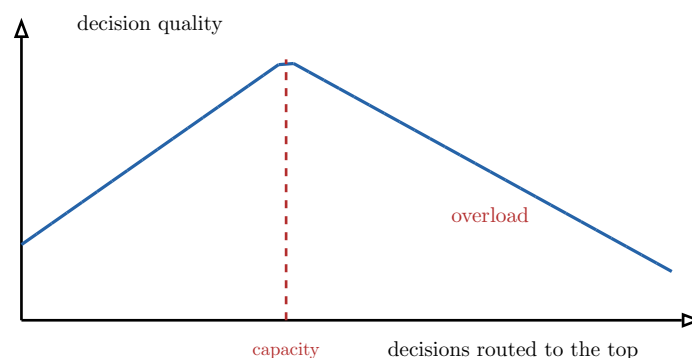


Figure 4: Hierarchy overload. Routing more decisions upward improves coherence up to the level’s attention capacity; beyond it, every decision degrades. The job is to keep the top below capacity by delegating decisions whose local information outweighs the value of central coherence.

4.3 Analysing the decision architecture

To analyse a firm’s decision-making architecture under attention constraints (Learning Outcome 4):

1. Identify **where** decisions are made and at what **volume**.
2. Identify where **hierarchy overload** is degrading decision quality — the saturated levels.
3. State what to **centralise** (decisions requiring coherence or carrying firm-wide consequence) and what to **delegate** (decisions whose local information outweighs the value of central coherence).

Common pitfall The instinct under pressure is to centralise more — “I’ll review everything until quality improves.” This is exactly backward once a level is saturated: pulling more decisions up degrades all of them. The fix for overload is almost always to *delegate* the decisions that do not need central coherence and protect executive attention for the few that do.

5 AI as cheap prediction: the Agrawal–Gans–Goldfarb framework

We now reach the course’s conceptual anchor. Everything before it treated capital and attention as the scarce resources. This chapter explains why, in the AI era, a third resource — human judgment — becomes the binding constraint, and how to decide which decisions to hand to machines and which to keep for people.

5.1 Prediction and judgment

The framework begins by splitting any decision into two components.

Definition 7 (Prediction and judgment) **Prediction** is the generation of information about what will happen — filling in missing information from the information you have. **Judgment** is the determination of the relative payoff of different outcomes — what the decision-maker values, and therefore what to do given the prediction. A decision

6 a prediction (what will happen?) combined with a judgment (what do we want, and what is it worth?).

Worked example — A loan decision, decomposed. Should Helios extend credit to a new logistics customer? The **prediction** is: will this customer default? — a pattern-recognition problem AI does superbly and cheaply. The **judgment** is: how do we weigh the profit from a paying customer against the loss from a defaulter, and how much do we value a long-term relationship over a single transaction? The machine predicts the default probability; a human decides what that probability is *worth*.

Artificial intelligence drives the cost of prediction toward zero. By basic economics, when the price of one input collapses, demand for its *complement* rises — and prediction’s complement is judgment. So the scarce, valuable, and increasingly decisive input becomes human judgment. This is the through-line of the entire course: *as AI makes prediction cheap, human judgment becomes the binding scarce asset*.

6.1 Classifying decisions

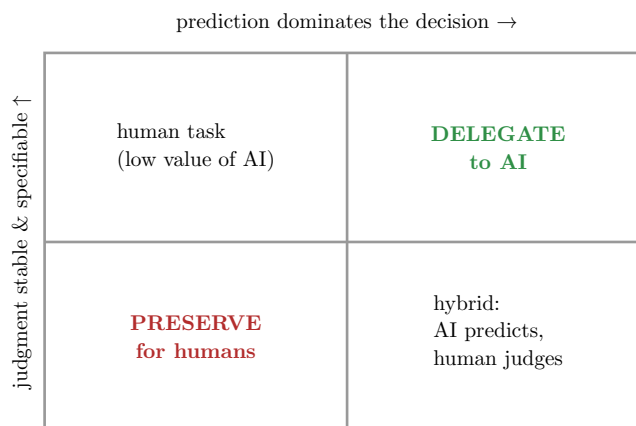


Figure 5: Classifying decisions. Delegate to AI where prediction dominates and the payoffs (judgment) are stable and specifiable. Preserve for humans where judgment dominates — contested, novel, or value-laden payoffs — even if the prediction is easy. Most real decisions are the hybrid bottom-right: machine predicts, human judges.

To apply the framework to a real organisation (Learning Outcome 5):

1. **Decompose** each decision into its prediction component and its judgment component.
2. **Delegate to AI** the decisions, or the parts of decisions, dominated by prediction whose judgment (the payoffs) is stable and specifiable.
3. **Preserve for human judgment** decisions whose payoffs are contested, novel, or value-laden — where the judgment component dominates.

6.2 Defending the human-judgment boundary

Here the framework becomes a leadership commitment, not just a classification exercise. Some decisions are *technically* delegable to AI — the prediction is easy and the payoffs look specifiable — yet should be preserved for human judgment anyway, as a **strategic-ethical commitment**. The defence rests on two grounds:

- **Competitive externalities** — consequences that fall *outside* the firm and outside the decision’s narrow payoff: labour displacement, winner-takes-most dynamics that concentrate power, and innovation-ecosystem effects that can hollow out the very markets the firm depends on. A decision that optimises the firm’s payoff while imposing these externalities may be one the board should own.
- **Accountability consequences** — decisions whose outcomes the board must answer for, to regulators, to society, to its own conscience. Accountability cannot be delegated to a model: a human must hold responsibility, which means a human must make — or at least own — the call.

Worked example — Helios automates hiring screens. Helios’s Data Services unit can deploy an AI to screen job applicants — the prediction (who will perform well?) is exactly what such systems do. Technically, it is delegable. Yet the board chooses to preserve human judgment over the final decision and the design of the screen, on two grounds: the competitive externality of large-scale labour-market effects and bias amplification, and the accountability consequence that Helios — not its vendor’s model — will answer for a discriminatory outcome. The decision stays on the human side of the boundary *because the board must remain answerable for it*, even though a machine could make it.

Common pitfall The boundary is not “wherever the AI is not yet good enough.” Capability and authority are different questions. A decision can be perfectly within AI’s predictive competence and still belong to humans because of the externalities it imposes or the accountability it carries. Drawing the boundary purely on technical capability is the characteristic error of the AI era — it cedes precisely the decisions a leader must own.

7 Organisational redesign for the AI era

If AI changes *where* prediction is cheap and judgment is scarce, then the decision structures built around the old cost of prediction are now misaligned. The leader’s task is to redesign them.

7.1 Reconfiguring decision structures

Much of a traditional organisation exists to gather, move, and process information — layers of analysts and middle managers whose job was, in effect, expensive prediction. When prediction becomes cheap, that scaffolding becomes overhead. Redesign reconfigures **which decisions are made where** and **which are delegated to machines versus preserved for humans** (Chapter 5).

Worked example — Helios flattens Logistics. Logistics once ran a layer of regional analysts who forecast demand and routed trucks — expensive prediction done by people. With an AI demand-and-routing engine, that prediction is near-free and instant. Helios reconfigures: the analyst layer shrinks, routing decisions are delegated to the machine, and the remaining managers move “up” the decision — from forecasting demand to *judging* which service-level

promises to make to which customers, a judgment the machine cannot make. The structure built for costly prediction is rebuilt around scarce judgment.

7.2 New sources of managerial leverage

Definition 8 (The shift in managerial leverage) When prediction is cheap, managerial leverage shifts away from the gathering and processing of information and toward the exercise of **judgment**: specifying payoffs, setting the objectives that predictions serve, deciding which decisions to preserve for humans, and integrating machine predictions into coherent action. The manager's value migrates from *knowing* to *judging*.

To propose an organisational redesign for a firm deploying AI at scale (Learning Outcome 6):

1. Specify **which decision structures to reconfigure** — which layers, committees, or roles existed to do prediction and can be rebuilt around judgment.
2. State **where managerial leverage shifts** as a result — what the remaining humans now do that the machine cannot.

Common pitfall Redesign is not “add an AI tool to each existing role and keep the structure.” Bolting cheap prediction onto a structure built for expensive prediction leaves the old bottlenecks and overhead in place and captures little of the gain. The redesign question is structural: which decisions move, which layers dissolve, and where human judgment is concentrated.

8 Measurable risk versus deep uncertainty

The later allocation decisions in this course are made under conditions the earlier finance courses largely assumed away. BUS33-3 and BUS43-3 taught you to optimise against a probability distribution. This chapter asks the prior question: *is there a distribution to optimise against at all?*

8.1 The distinction, and the rule it dictates

Definition 9 (Measurable risk and deep uncertainty) **Measurable risk** is a situation in which the possible outcomes and their probabilities are known or estimable — you can write down a distribution. **Deep uncertainty** (radical ambiguity) is a situation in which the outcomes, their probabilities, or both *cannot* be specified — there is no reliable distribution to write down.

The allocation rule depends entirely on which situation you are in:

- under **measurable risk**, **optimisation** is appropriate — maximise expected value given the known distribution (the toolkit of BUS33-3/BUS43-3);
- under **deep uncertainty**, optimisation is unsafe, because the distribution it optimises against does not exist; **robustness** is appropriate — choose actions that perform *acceptably across the range* of possible futures rather than *optimally against one* assumed future.

Common pitfall The most dangerous and most common error is to treat deep uncertainty as if it were measurable risk — to invent a probability distribution and optimise against it,

producing a precise answer to the wrong problem. A confident expected-value calculation built on fabricated probabilities is more dangerous than admitted ignorance, because it *feels* rigorous. The first step is always to classify the situation, not to compute.

8.2 Robust versus optimised strategies

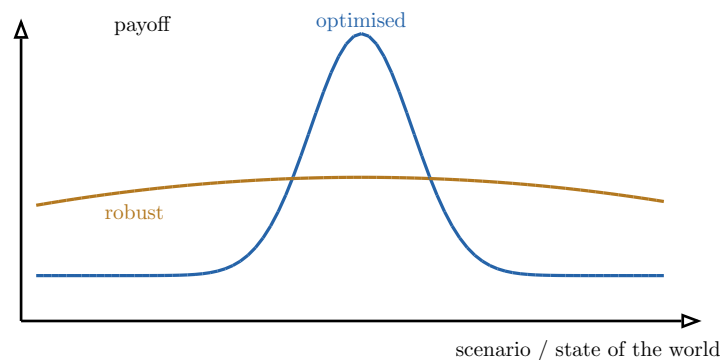


Figure 6: Optimised versus robust. The optimised strategy (blue) wins big in the one scenario it was tuned for and collapses elsewhere. The robust strategy (orange) never wins big but performs acceptably across every scenario. Under turbulence — when you cannot say which scenario will occur — robust dominates.

To compare them for a firm facing turbulent conditions (Learning Outcome 9), name the scenarios under which each outperforms:

- the **optimised** strategy outperforms when its *assumed scenario actually materialises* — it is the right choice precisely when you genuinely know the distribution;
- the **robust** strategy outperforms when the future *departs from any single assumed scenario* — i.e. under genuine turbulence, where being approximately right everywhere beats being exactly right in a world that did not arrive.

9 Real-options reasoning

Robustness handles uncertainty by performing acceptably everywhere. Real options handle it differently — by *preserving the right to act later*, once the uncertainty resolves. This is the forward-looking allocation tool the ROIC compass could not provide, and it is what makes irreversibility the central concept of the chapter.

9.1 The value of flexibility

Definition 10 (Real-options reasoning) Real-options reasoning (introduced as intuition in BUS43-3) treats a strategic commitment as analogous to a financial option: the *right, but not the obligation*, to act later — to delay, expand, contract, or abandon. Flexibility has value because it lets the firm act on information that arrives *after* the initial decision. The more uncertain the environment and the more irreversible the commitment, the more that flexibility is worth.

Worked example — Helios and the new market. Helios is considering entering a new geography. It can **commit now** — build a full network for €400m, capturing first-mover scale — or **preserve optionality** — open a small pilot for €40m, keeping the right to scale later once demand is proven. If demand is uncertain, the pilot is worth more than its lower

expected payoff suggests, because it buys the *option* to commit fully only in the world where commitment pays. The €40m pilot is the price of an option on the €400m decision.

9.2 Commit versus preserve optionality

Definition 11 (The commit-versus-preserve-optionality threshold) The point at which the value of committing now — pre-empting a rival, capturing scale or learning, ending costly delay — exceeds the value of waiting to preserve flexibility. *Below* the threshold, preserve optionality; *above* it, commit. The threshold rises with uncertainty and irreversibility (which favour waiting) and falls with the value of pre-emption and the cost of delay (which favour committing).

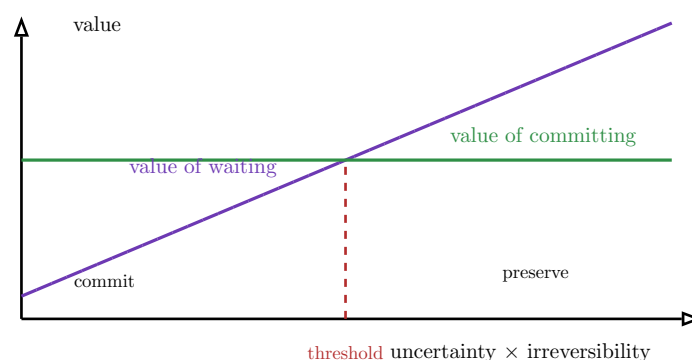


Figure 7: The commit-versus-preserve threshold. The value of waiting rises with uncertainty and irreversibility; the value of committing (pre-emption, scale) is roughly fixed. Left of the threshold, commit; right of it, preserve optionality. Irreversibility pushes the whole “wait” line up.

9.3 Irreversibility and explicit exit conditions

Definition 12 (The cost of irreversibility) An irreversible commitment forecloses options: once made, it cannot be costlessly undone. The cost of irreversibility is the value of the flexibility surrendered by committing — a real cost that a naïve NPV, which ignores the abandoned options, leaves out entirely.

To apply real-options reasoning to an irreversible strategic commitment (Learning Outcome 8):

1. Name the **value of flexibility** the commitment would consume.
2. State the **commit-versus-preserve-optionality threshold** — the condition under which committing now beats waiting.
3. Specify **explicit exit conditions** — the observable conditions that would invalidate the commitment and trigger withdrawal, fixed *in advance*.

Common pitfall Exit conditions must be written down *before* committing, in observable terms (“if cumulative volume is below X by quarter Y, we withdraw”). Set after the fact, they are rationalised away: every disappointing result gets reinterpreted as “just early,” and the firm rides a failing commitment to the bottom. This is the sunk-cost trap institutionalised. Pre-committed, observable exit conditions are the discipline that converts irreversibility from a trap into a managed risk — and the same discipline the capstone audit applies to its own recommendations.

10 The strategic allocation audit

Everything converges here. The terminal deliverable is a strategic allocation audit of a real organisation, and its purpose is to prove that the tools of this course are not separate chapters: a real allocation decision touches all of them at once, and they must be made *mutually coherent*.

10.1 Coherence is the deliverable

Definition 13 (The strategic allocation audit) An audit of a real organisation that tests the **coherence** among four things:

1. its **declared strategy** — the competitive logic (BUS33-2);
2. its **capital allocation** — where capital is actually deployed, judged on ROIC versus WACC (Chapter 1);
3. its **organisational design** — where decisions are made, under attention constraints and AI deployment (Chapters 4–6);
4. its **make/buy/partner decisions** — corporate scope (Chapters 2–3).

The organisation is creating value where these four cohere and destroying it where they conflict.

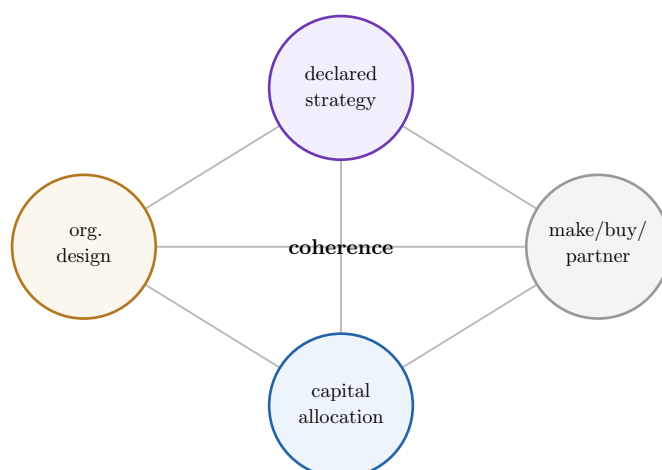


Figure 8: The coherence test. The audit checks every edge: does the capital allocation back the declared strategy? Does the org design support it? Do the make/buy/partner choices cohere with both? Value is destroyed at any edge where two corners contradict each other.

Worked example — Helios, audited. The audit only becomes a verdict when one diagnosis runs through all four corners. Helios’s **declared strategy** is focus on high-return logistics and industrial niches. Yet its **capital allocation** keeps funding Data Services at 4% ROIC — a direct contradiction of the focus strategy. Its **organisational design** centralises every €5m decision in one saturated committee (Chapter 4) — incoherent with a strategy that depends on local logistics responsiveness. And its **make/buy/partner** choice to rent cold-chain capability undercuts the very differentiation the strategy claims. Four corners, four contradictions: the audit’s verdict is that Helios’s declared strategy and its actual commitments have diverged, and it names the specific redeployments that would restore coherence.

10.2 What the deliverable must carry

The audit produces **quantified recommendations** — grounded in the financial measures the prerequisites established (ROIC versus WACC, the value transferred in a deal, the spread redeployment would create), not assertions — and **explicit invalidation conditions**: the

observable conditions under which each recommendation would no longer hold. This is the exit-condition discipline of Chapter 8 turned on the audit itself. It is produced and defended before a jury.

Common pitfall The audit fails not when a single number is wrong but when the four corners are analysed well *in isolation* and never tested against each other. A beautiful ROIC analysis, a sharp org-design critique, and a clean scope review that never confront one another miss the entire point — coherence *is* the deliverable. Juries, like real boards, probe exactly the seams between the corners. And a recommendation with no invalidation condition is not a recommendation; it is a hope.

The through-line. Prediction is cheap; the binding scarce assets are *capital*, *attention*, and — above all — *human judgment*. Strategic allocation is the discipline of committing those scarce assets coherently. Deploy capital only where ROIC beats WACC, and never confuse growth with value (Ch. 1). Choose make/buy/partner and corporate scope on coherence, flexibility, and exposure — not cost — and exercise divestiture discipline (Ch. 2). Treat M&A as the instrument that most reliably destroys value, and judge every deal by mechanism and counterfactual (Ch. 3). Allocate scarce executive attention as deliberately as capital, delegating to escape hierarchy overload (Ch. 4). As AI makes prediction cheap, route prediction to machines but preserve for human judgment the decisions whose externalities and accountability a board must own (Ch. 5), and redesign the organisation around scarce judgment rather than cheap prediction (Ch. 6). Classify each decision as measurable risk or deep uncertainty, optimising only when a distribution truly exists and choosing robustness when it does not (Ch. 7). Use real options to value flexibility, commit irreversibly only above the threshold, and write exit conditions in advance (Ch. 8). Finally, prove it all coheres — strategy, capital, organisation, and scope as one designed whole — in the strategic allocation audit (Ch. 9). In the leader's chair, these were never separate decisions.